

IT & User Use cases to monitor



IT & User use cases to monitor

IT Use cases from a network:

1. User authentication
2. Web Browsing activity
3. Email communication
4. File transfer
5. Remote Connections
6. Remote Sessions
7. MS Teams activity monitoring

Labs

- Students to analyze the given event viewer logs and investigate the login logout activity
- Analyze the given proxy logs to identify the suspicious activity
- Analyze the given proxy logs to identify access to prohibited social media sites

Network Monitoring – User Authentication

- **Client–Server communication**

- Endpoint → Domain Controller / IdP (AD, LDAP, Azure AD)
- VPN gateway → AAA server (RADIUS)

- **Data flow**

- Kerberos tickets
- NTLM challenges
- LDAP bind requests
- SAML/OAuth tokens

- **Ports / Protocols / Services**

- Kerberos: TCP/UDP 88
- LDAP: 389
- LDAPS: 636
- RADIUS: 1812
- TACACS+: 49

- **Systems & devices involved**

- End user devices
- Domain Controllers
- IdP (Azure AD/Okta)
- VPN concentrator
- NAC appliances

- **Security controls (L3–L7)**

- L3: IP ACLs
- L4: Port filtering
- L5: Session timeouts
- L6: TLS encryption
- L7: MFA, conditional access

Logs to collect

- Windows Security logs (Event IDs 4624, 4625)
- RADIUS logs
- VPN authentication logs
- Azure AD sign-in logs

Network Monitoring – Web Browsing Activity

- **Client–Server communication**

- User device → Proxy / Internet gateway → Web server

- **Data flow**

- HTTP GET/POST
- HTTPS CONNECT tunnels
- DNS resolution

- **Ports / Protocols / Services**

- HTTP: 80
- HTTPS: 443
- DNS: 53
- Proxy: 3128 (Squid)

- **Applications / systems**

- Browsers
- Proxy servers
- Secure web gateways
- CDN endpoints

- **Security controls**

- L3: Firewall routing
- L4: port restriction
- L7: URL filtering, SSL inspection, CASB

Logs required

- Squid access.log
- Firewall logs
- DNS query logs
- Secure Web Gateway logs

Network Monitoring – Email Communication

- **Client–Server communication**

- Outlook → Exchange/M365
- SMTP relay → internet mail servers

- **Data flow**

- SMTP send
- IMAP/POP retrieval
- Attachments

- **Ports / Protocols**

- SMTP: 25
- SMTPS: 465
- Submission: 587
- IMAP: 143 / 993
- POP3: 110 / 995

- **Systems involved**

- Mail servers
- Spam gateways
- DLP engines
- Endpoint clients

- **Security controls**

- SPF, DKIM, DMARC
- Email encryption
- Attachment sandboxing

Logs required

- Exchange message tracking logs
- SMTP gateway logs
- DLP alerts
- Email security platform logs

Network Monitoring – File Transfer

- **Client–Server communication**

- Endpoint ↔ File servers / NAS / cloud storage

- **Data flow**

- Upload/download
- SMB shares
- SFTP sessions

- **Ports / Protocols**

- SMB: 445
- NFS: 2049
- FTP: 21
- SFTP: 22
- HTTPS cloud transfer: 443

- **Systems involved**

- NAS
- SharePoint
- OneDrive
- FTP/SFTP servers

- **Security controls**

- Access control lists
- DLP
- Encryption in transit

- **Logs required**

- File server audit logs
- SFTP logs
- Cloud storage access logs
- NetFlow bandwidth data

Network Monitoring – Remote Connections

- **Client–Server communication**

- Remote user → VPN gateway → internal systems

- **Data flow**

- Encrypted tunnel
- Application access inside VPN

- **Ports / Protocols**

- IPsec: UDP 500/4500
- SSL VPN: 443
- WireGuard: 51820

- **Systems involved**

- VPN concentrator
- Endpoint
- Internal application servers

- **Security controls**

- MFA
- Endpoint posture check
- Device compliance

- **Logs required**

- VPN session logs
- Authentication logs
- Endpoint compliance logs

Network Monitoring – Remote Sessions

- **Client–Server communication**

- Admin workstation → servers/network devices

- **Data flow**

- Command execution
- Screen sessions

- **Ports / Protocols**

- SSH: 22
- RDP: 3389
- VNC: 5900
- WinRM: 5985/5986

- **Systems involved**

- Jump servers
- Privileged Access Management (PAM)
- Servers

- **Security controls**

- Session recording
- Privileged identity control
- Bastion hosts

- **Logs required**

- RDP logs
- SSH audit logs
- PAM logs
- Command history logs

Network Monitoring – Microsoft Teams Activity Monitoring

- **Client–Server communication**

- Teams client → Microsoft 365 cloud

- **Data flow**

- Video streams
- Chat messages
- File sharing

- **Ports / Protocols**

- HTTPS: 443
- STUN/TURN: UDP 3478–3481
- Media: UDP 50000–50059

- **Applications/systems**

- Teams client
- Microsoft cloud edge nodes
- Identity services

- **Security controls**

- Conditional access
- DLP policies
- Meeting controls

- **Logs required**

- M365 audit logs
- Teams activity reports
- Azure AD logs
- Defender for Cloud Apps telemetry

User Authentication Monitoring

- **Client–Server communication**

- Endpoint → Domain Controller / Identity Provider
- Endpoint → VPN Gateway → AAA server
- Application → SSO platform
- Capture:
 - Source/destination IP
 - Username
 - Session initiation time
 - Device identity

- **Data flow**

- Kerberos ticket exchange
- NTLM challenge/response
- LDAP bind packets
- Token issuance (OAuth/SAML)

- **Ports, Protocols, Services**

- TCP/UDP 88 — Kerberos
- TCP 389/636 — LDAP/LDAPS
- TCP 443 — SSO/Identity federation
- UDP 1812 — RADIUS

- **Applications, systems, devices**

- Endpoint (Windows/Linux/Mobile)
- Domain Controller / Azure AD
- VPN appliance
- NAC systems

User Authentication Monitoring – Security Controls

Layer	Controls
L3	IP whitelisting
L4	Port restrictions
L5	Session control
L6	TLS encryption
L7	MFA, IAM policies

Web Browsing Activity Monitoring

- **Client–Server communication**

- Endpoint → Proxy → Internet web server
- Endpoint → DNS resolver

- **Data flow**

- HTTP requests/responses
- HTTPS TLS sessions
- DNS queries

- **Ports / Protocols / Services**

- HTTP: 80
- HTTPS: 443
- DNS: 53
- Proxy: 3128

- **Applications, systems, devices**

- Browsers
- Proxy servers (Squid)
- SWG/CASB
- CDN/web servers



Web Browsing Activity Monitoring – Security Controls

Layer	Controls
L3	Firewall routing
L4	Port filtering
L6	SSL inspection
L7	URL filtering

Email Communication Monitoring

- **Client–Server communication**

- Mail client → mail server
- Mail server → internet SMTP relay

- **Data flow**

- SMTP send
- IMAP/POP retrieval
- Attachments

- **Ports / Protocols**

- SMTP 25/587
- SMTPS 465
- IMAP 143/993
- POP3 110/995

- **Applications, systems, devices**

- Outlook/Thunderbird
- Exchange/M365
- Email gateways
- Spam filters

Email Communication Monitoring - Security controls

Layer	Controls
L3	Email routing control
L4	Mail server port access
L6	TLS encryption
L7	Anti-spam, DLP

File Transfer Monitoring

- **Client–Server communication**

- Endpoint ↔ NAS/File server/cloud storage

- **Data flow**

- File upload/download
 - SMB transactions
 - SFTP transfers

- **Ports / Protocols**

- SMB 445
 - FTP 21
 - SFTP 22
 - HTTPS 443

- **Applications, systems, devices**

- NAS
 - SharePoint
 - OneDrive
 - FTP servers

File Transfer Monitoring - Security controls

Layer	Controls
L3	Network segmentation
L4	Transfer ports
L6	Encryption
L7	DLP, file inspection

Remote Connections Monitoring

- **Client–Server communication**

- Remote endpoint → VPN → enterprise network

- **Data flow**

- Encrypted tunnel packets
- Internal application access

- **Ports / Protocols**

- IPsec UDP 500/4500
- SSL VPN 443
- WireGuard 51820

- **Applications/systems/devices**

- VPN gateway
- Endpoint device
- Internal servers

Remote Connections Monitoring - Security controls

Layer	Controls
L3	Tunnel routing
L4	VPN port control
L6	Encryption
L7	Endpoint posture validation

Remote Sessions Monitoring

- **Client–Server communication**

- Admin workstation → server/jump host

- **Data flow**

- SSH commands
- RDP screen sessions

- **Ports / Protocols**

- SSH 22
- RDP 3389
- WinRM 5985/5986

- **Applications, systems, devices**

- Jump server
- PAM solution
- Managed servers

- **Security Controls**

Layer	Controls
L3	Admin network isolation
L4	Port restriction
L6	Secure session encryption
L7	Privileged session recording

Microsoft Teams Activity Monitoring

- **Client–Server communication**

- Teams client → Microsoft cloud services

- **Data flow**

- Video/audio packets
- Chat messages
- File sharing

- **Ports / Protocols**

- HTTPS 443
- UDP 3478–3481
- UDP media streams 50000+

- **Applications/systems/devices**

- Teams desktop/mobile app
- Microsoft 365 cloud
- Identity services

- **Security controls**

Layer	Controls
L3	Cloud traffic routing
L4	Media ports
L6	TLS encryption
L7	Conditional access, DLP

User Authentication

In Azure

Date : Last 24 hours

Show dates as : Local

+ Add filters

User sign-ins (interactive)

User sign-ins (non-interactive)

Service principal sign-ins

Managed identity sign-ins

Date	Request ID	User	Application	Status	IP address	Location
8/16/2023, 9:18:26 AM	0000111-aaaaa-2222	Izaak Schmitz	Microsoft App Acces...	Success	10.0.1.18	Sao Paulo, Sao Paulo, BR
8/16/2023, 9:17:46 AM	111000-bbbb00-221...	Shokat Nadeem	Microsoft App Acces...	Success	172.16.8.1	Philadelphia, Pennsylv...
8/16/2023, 9:17:29 AM	11100aa-111aaaa-b...	Emma Blug	Medeina Portal Dev	Success	192.168.5.1	Redmond, Washington,...
8/16/2023, 9:17:26 AM	aaaa1111-000111-aa...	Kasper Graf	Medeina Portal Dev	Interrupted	192.168.200.4	Redmond, Washington,...



User Authentication

In PowerShell

Cmdlet:

Get-WinEvent -

FilterHashtable

@{LogName='Security';

Id=4625} | ft -AutoSize -

Wrap

```
Administrator: Windows PowerShell
> Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625} | ft -AutoSize -Wrap

ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated          Id LevelDisplayName Message
-----
15-Dec-25 9:39:52 AM 4625 Information    An account failed to log on.

Subject:
  Security ID:          S-1-5-18
  Account Name:         WORKSTATION$
  Account Domain:      WORKGROUP
  Logon ID:             0x3E7

Logon Type:          2

Account For Which Logon Failed:
  Security ID:          S-1-0-0
  Account Name:         Jeetu
  Account Domain:      WORKSTATION

Failure Information:
  Failure Reason:       Unknown user name or bad password.

Status:              0xC000006D

Sub Status:          0xC000006A
```

Capturing Network Connections (Near Real-Time)

1. View Active Web Connections:

- Cmdlet: `Get-NetTCPConnection Where-Object { $_.RemotePort -eq 80 -or $_.RemotePort -eq 443 }`

```
1  Get-NetTCPConnection |
2  Where-Object { $_.RemotePort -eq 80 -or $_.RemotePort -eq 443 }
```

PS C:\WINDOWS\system32> Get-NetTCPConnection |
Where-Object { \$_.RemotePort -eq 80 -or \$_.RemotePort -eq 443 }

LocalAddress	LocalPort	RemoteAddress
192.168.1.8	65342	172.64.155.209
192.168.1.8	63579	172.64.155.209
192.168.1.8	62464	104.18.39.21
192.168.1.8	61662	4.213.25.241
192.168.1.8	57595	150.171.84.40
192.168.1.8	54475	172.64.155.209
192.168.1.8	53214	18.237.153.92
192.168.1.8	52793	20.189.173.12



Capturing Network Connections (Near Real-Time)

2. Map Connections to Processes:

- Cmdlet:

Get-NetTCPConnection |

Where-Object { \$_.RemotePort -eq 443 } |

Select LocalAddress, RemoteAddress, OwningProcess

LocalAddress	RemoteAddress	OwningProcess
192.168.1.8	52.110.21.101	0
192.168.1.8	52.168.112.67	7996
192.168.1.8	20.190.145.142	0
192.168.1.8	20.190.145.142	0
192.168.1.8	172.64.155.209	19292
192.168.1.8	104.18.39.21	19292
192.168.1.8	4.213.25.241	6032
192.168.1.8	18.237.153.92	3716
192.168.1.8	20.189.173.12	0



Capturing Network Connections (Near Real-Time)

3. Resolve process:

- Cmdlet:

Get-Process -Id <PID>

```
Untitled1.ps1* X
1  cls
2  Get-Process -Id 7996
3
```

Handles	NPM(K)	PM(K)	WS(K)	CPU(s)	Id	SI	ProcessName
2465	109	576560	697212	211.73	7996	1	POWERPNT

```
Untitled1.ps1* X
1  cls
2  Get-Process -Id 19292
3
```

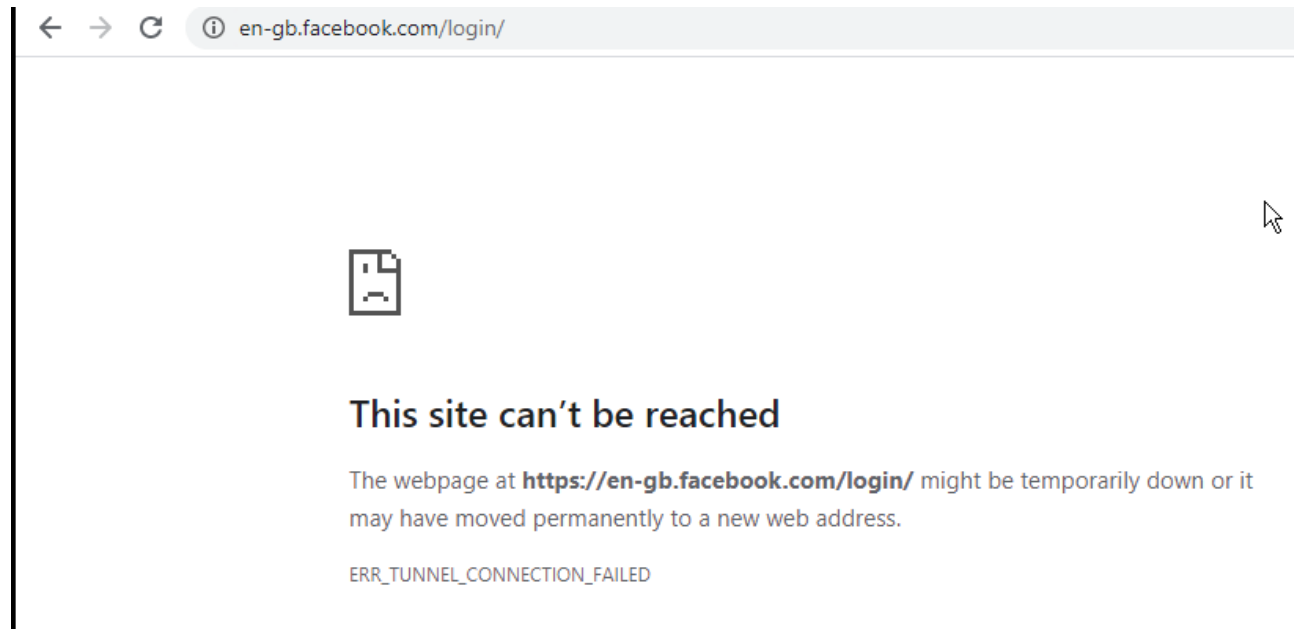
Handles	NPM(K)	PM(K)	WS(K)	CPU(s)	Id	SI	ProcessName
495	33	31820	68868	25.28	19292	1	chrome



Capturing Network Connections (using SQUID)

```
root@svr:~  
[root@svr ~]# grep ziyad /var/log/squid/access.log  
1587186645.776 35645 192.168.6.137 TCP_MISS/200 2069852 CONNECT ziyad-bhombal.web.app:  
443 - HIER_DIRECT/151.101.1.195 -  
1587186822.900 173821 192.168.6.137 TCP_MISS/200 2067664 CONNECT ziyad-bhombal.web.app:  
443 - HIER_DIRECT/151.101.1.195 -  
[root@svr ~]#
```

Monitoring and Blocking
unwanted websites



Monitoring Email communication

- In cybersecurity, email monitoring does not mean reading every email.
- It means monitoring:
 - Authentication and access to mailboxes
 - Email flow (sender, recipient, subject, attachments, URLs)
 - Malicious indicators (phishing, malware, BEC)
 - Policy violations (DLP, data exfiltration)
 - Administrative and configuration changes

Area	Tool
Email Flow	Message Trace
Phishing	Defender O365
Mailbox Access	Audit Logs
Insider Threat	Inbox Rules
DLP	Compliance Portal
SIEM	Microsoft Sentinel



Monitoring Remote Connections

Monitoring failed
connection

```
1 cls
2 Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625} | `
3 Format-Table -AutoSize -Wrap
4
```

```
ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated          Id LevelDisplayName Message
-----
15-Dec-25 9:39:52 AM 4625 Information An account failed to log on.

Subject:
Security ID:        S-1-5-18
Account Name:       WORKSTATION$
Account Domain:     WORKGROUP
Logon ID:           0x3E7

Logon Type:         2

Account For which Logon Failed:
Security ID:        S-1-0-0
Account Name:       Jeetu
Account Domain:     WORKSTATION

Failure Information:
Failure Reason:     Unknown user name or bad password.
Status:            0xC000006D
Sub Status:        0xC000006A

Process Information:
Caller Process ID:  0x8b4
Caller Process Name: C:\windows\system32\svchost.exe
```



Monitoring Remote Connections – Active users

```
1  cls
2  Write-Host "Cmd: qwinsta" -ForegroundColor Yellow
3  qwinsta
4
5  #or
6  Write-Host "Cmd: query user" -ForegroundColor Yellow
7  query user
```

Cmd: qwinsta

SESSIONNAME	USERNAME	ID	STATE	TYPE	DEVICE
services		0	Disc		
>console	Jeetu	1	Active		

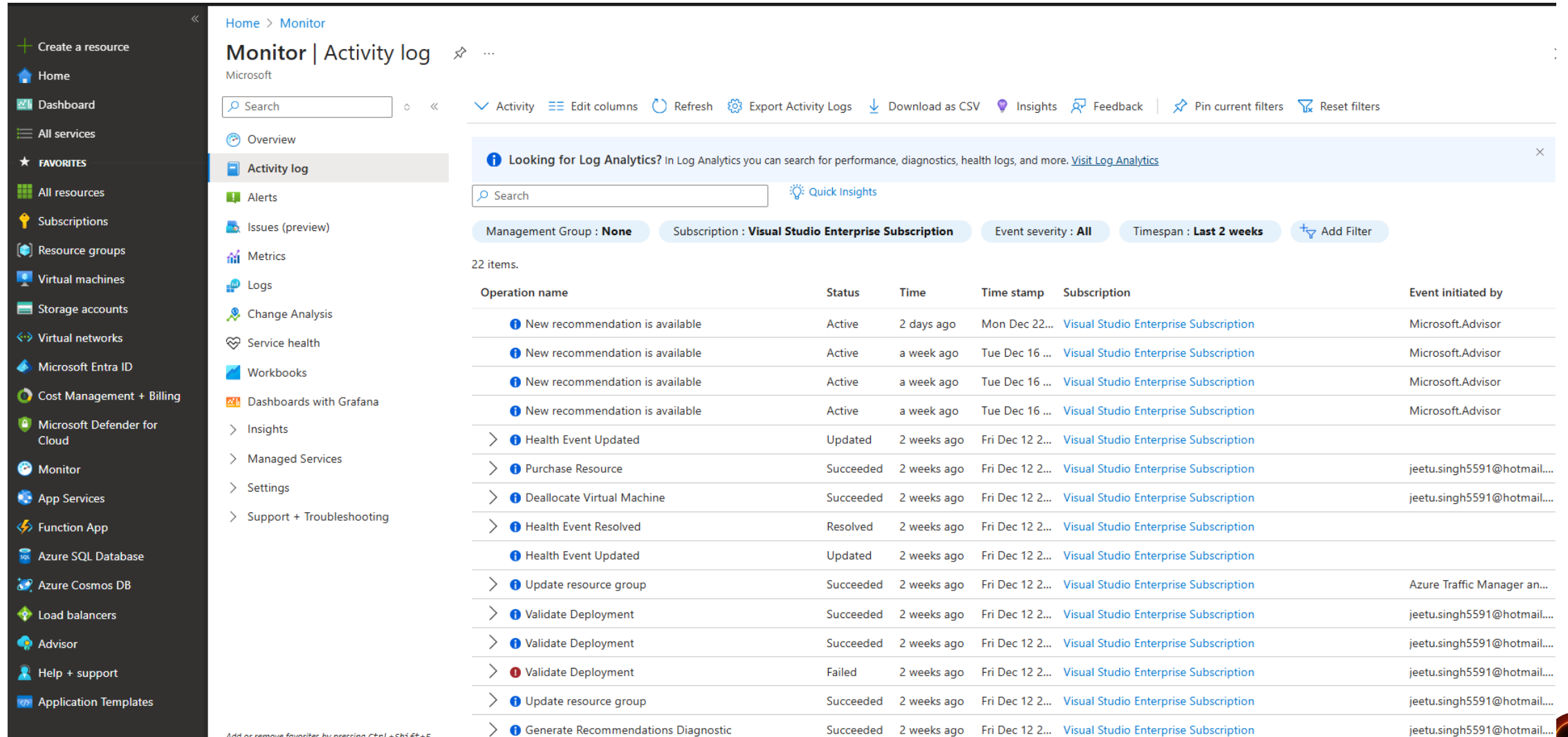
Cmd: query user

USERNAME	SESSIONNAME	ID	STATE	IDLE TIME	LOGON TIME
>jeetu	console	1	Active	none	24-Dec-25 9:25 AM

PS C:\WINDOWS\system32>



Monitoring Remote Connections – Azure



The screenshot displays the Azure Monitor Activity log interface. The left sidebar contains navigation options such as 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main content area is titled 'Monitor | Activity log' and shows a list of 22 items. The table columns are 'Operation name', 'Status', 'Time', 'Time stamp', 'Subscription', and 'Event initiated by'. The events include 'New recommendation is available', 'Health Event Updated', 'Purchase Resource', 'Deallocate Virtual Machine', 'Health Event Resolved', 'Health Event Updated', 'Update resource group', 'Validate Deployment', and 'Generate Recommendations Diagnostic'.

Home > Monitor

Monitor | Activity log

Microsoft

Search

Activity Edit columns Refresh Export Activity Logs Download as CSV Insights Feedback Pin current filters Reset filters

Overview

Activity log

Alerts

Issues (preview)

Metrics

Logs

Change Analysis

Service health

Workbooks

Dashboards with Grafana

Insights

Managed Services

Settings

Support + Troubleshooting

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Search Quick Insights

Management Group : None Subscription : Visual Studio Enterprise Subscription Event severity : All Timespan : Last 2 weeks Add Filter

22 items.

Operation name	Status	Time	Time stamp	Subscription	Event initiated by
New recommendation is available	Active	2 days ago	Mon Dec 22...	Visual Studio Enterprise Subscription	Microsoft.Advisor
New recommendation is available	Active	a week ago	Tue Dec 16 ...	Visual Studio Enterprise Subscription	Microsoft.Advisor
New recommendation is available	Active	a week ago	Tue Dec 16 ...	Visual Studio Enterprise Subscription	Microsoft.Advisor
New recommendation is available	Active	a week ago	Tue Dec 16 ...	Visual Studio Enterprise Subscription	Microsoft.Advisor
Health Event Updated	Updated	2 weeks ago	Fri Dec 12 2...	Visual Studio Enterprise Subscription	
Purchase Resource	Succeeded	2 weeks ago	Fri Dec 12 2...	Visual Studio Enterprise Subscription	jeetu.singh5591@hotmail...
Deallocate Virtual Machine	Succeeded	2 weeks ago	Fri Dec 12 2...	Visual Studio Enterprise Subscription	jeetu.singh5591@hotmail...
Health Event Resolved	Resolved	2 weeks ago	Fri Dec 12 2...	Visual Studio Enterprise Subscription	
Health Event Updated	Updated	2 weeks ago	Fri Dec 12 2...	Visual Studio Enterprise Subscription	
Update resource group	Succeeded	2 weeks ago	Fri Dec 12 2...	Visual Studio Enterprise Subscription	Azure Traffic Manager an...
Validate Deployment	Succeeded	2 weeks ago	Fri Dec 12 2...	Visual Studio Enterprise Subscription	jeetu.singh5591@hotmail...
Validate Deployment	Succeeded	2 weeks ago	Fri Dec 12 2...	Visual Studio Enterprise Subscription	jeetu.singh5591@hotmail...
Validate Deployment	Failed	2 weeks ago	Fri Dec 12 2...	Visual Studio Enterprise Subscription	jeetu.singh5591@hotmail...
Update resource group	Succeeded	2 weeks ago	Fri Dec 12 2...	Visual Studio Enterprise Subscription	jeetu.singh5591@hotmail...
Generate Recommendations Diagnostic	Succeeded	2 weeks ago	Fri Dec 12 2...	Visual Studio Enterprise Subscription	jeetu.singh5591@hotmail...

Add or remove favorites by pressing Ctrl+Shift+F



Network-Level Monitoring - Active Connections

- `Get-NetTCPConnection | Where-Object {$_.State -eq "Established"}`

```
PS C:\Users\Jeetu> Get-NetTCPConnection | Where-Object {$_.State -eq "Established"}
```

LocalAddress	LocalPort	RemoteAddress	RemotePort	State	AppliedSetting
192.168.1.8	61478	40.126.18.33	443	Established	Internet
192.168.1.8	61475	104.18.32.47	443	Established	Internet
192.168.1.8	60444	172.64.148.235	443	Established	Internet
192.168.1.8	59635	142.251.12.188	5228	Established	Internet
192.168.1.8	59034	163.70.143.60	443	Established	Internet
192.168.1.8	53447	4.213.25.242	443	Established	Internet
127.0.0.1	49674	127.0.0.1	49673	Established	Internet
127.0.0.1	49673	127.0.0.1	49674	Established	Internet







That's all Folks!